MICROBURBS HACKATHON · TASK 4 · TEAM-OPAL

Would you let one flat vote ten times?

An evidence-weighting AVM: each flat's own past rent, its neighbours in the same building (one vote each), and the view from its floor.

$45
typical miss, 2,898 held-out Jun–Jul adverts

77%
within $100 a week

0.88
R^2, vs 0.82 for ML trees

6 / 6
walk-forward origins where each core rule wins

Organisers' reference on the scored Aug–Sep adverts: simple estimate $100, Microburbs rental valuation $86. Different adverts.

THE PROBLEM

Same building, same bedrooms — up to $500 apart

45 Shelley St, Sydney · one-bedroom flats

floor	water in sight	asking rent
2–3	0% of directions	$700 – $875
4–5	3–17%	$685 – $850
6–7	**25–33%**	**$875 – $1,200**

Area averages can't see it.

Government bond data (postcode × bedrooms): $100 typical miss.

The flat's own evidence can.

Its past rent and its neighbours' rents already price floor, view, noise and renovation.

So the question becomes: which evidence do we trust?

Use the strongest evidence each flat has — and weigh it

Own history	the flat's latest advert with the same bedrooms, under 3 years old, moved to today by a repeat-rent index	**74% of held-out adverts · $40 error**
Building neighbours	same building and bedrooms, one vote per flat however often it relisted, shrunk to the suburb when few	**26% (with the ladder) · $60 error, was $99**
Suburb ladder	suburb × bedrooms × parking medians, index-adjusted, ≥10 adverts	**last resort**
Fine-tuning	LiDAR floor-view difference vs the neighbours; parking premium; $5 asking grid	**applies to every estimate**

Every estimate can name its evidence: which advert, which neighbours, which adjustment.

The features carry the accuracy

feature	how it is built (from adverts before the cut-off)	what it did
repeat-rent index	pairs of adverts of the same flat; 60-day lettings; monthly log index	turns any past rent into today's
own history (same bedrooms)	latest advert with matching bedrooms	bedroom mismatch: error × 2.4
stale flag	own advert older than 36 months not trusted	$53 → $46.5 (window A)
one-vote building comps	one signal per neighbour flat, target out, n/(n+0.5)	no-history $99 → $60
parking premium	with vs without, suburb × bedrooms, median 8%	second output column
LiDAR floor view	water / park / roofline per floor vs neighbours	R^2 0.877 → 0.880
asking-rent grid	84% of rents end in $0	$46.5 → $45

Kept out to avoid leakage: first_ever_listed, attrs_last_seen, API leased flag, rent-map; Airbnb snapshot (scraped after the cut-off).

Why this AVM, not a regression or a black box



within $100 · R^2

regression 58% · 0.71

regression + our features 66% · 0.78

ML trees 78% · 0.82

ours 77% · 0.88

A regression without history misses by $83; with our features, $69 — fixed linear weights cannot switch evidence on and off per flat.

ML trees only catch up when fed our engineered evidence — the features, not the model, do the work.

Same accuracy, but ours has the best R^2 (fewer big misses), needs no training lag, and explains every estimate.

Every rule tested — none taken on faith

rule	origins where it wins	mean effect
building comps (vs none)	6 / 6	$57.5 → $50.0
one vote (vs every advert)	6 / 6	within $100 75.6% → 76.0%
stale rule 36 m (vs none)	6 / 6	$55.8 → $50.0
shrinkage k = 0.5 (vs 2, 5)	6 / 6	best on every origin
no uplift after cut-off (vs trend)	4 / 6	$51.7 → $50.0
LiDAR floor view (vs none)	3 / 6 (2 worse)	small, positive

Fixed constants

11 constants × 2 alternatives on 4 origins: most alternatives tie or lose.

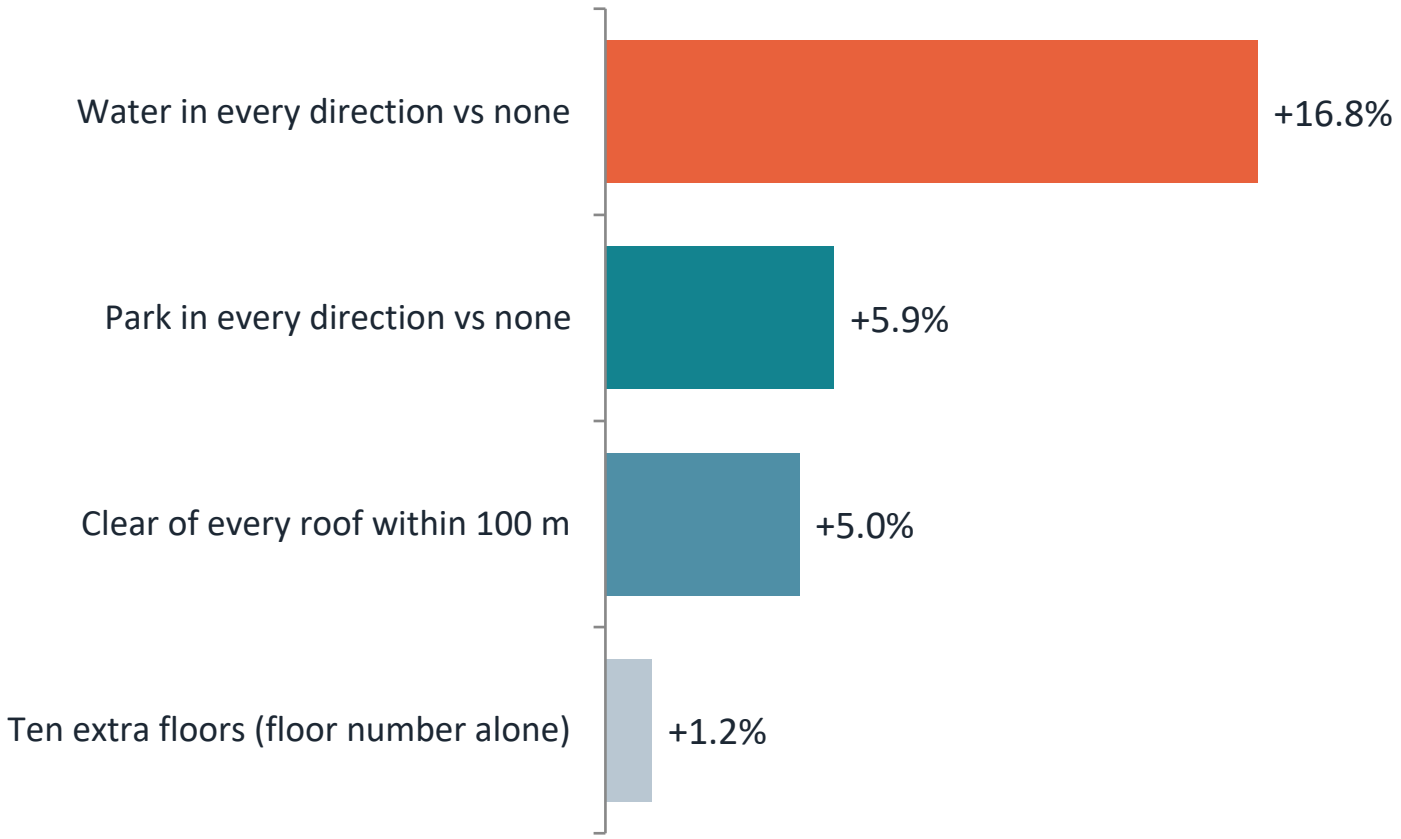
Fixed before any real data

Committed 09:12–09:19; API key 11:40, official files 12:15.

Honest next steps

Walk-forward found 24-month staleness, stronger suburb shrinkage and a $150–$5,000 filter ~$1–2 better. Not applied: chosen after the freeze.

It's the view, not the floor



Same building, same bedrooms

22,905 flats in 2,912 groups; building location, age and quality cancel out.

Two surveys (2013, 2020), 5.6 GB, 4,399 buildings, 29,970 floors, 36 directions to 1.5 km.

Building-level layers (heat, canopy, flight path, Airbnb) double-counted the neighbours' rents: tested, dropped.

A rent you can check, flat by flat

Tenants

is this asking rent high for this floor of this building?

Landlords

price against the flats around you, not the suburb

Investors

likely rent before you buy, with its evidence

github.com/HollyHsu0808/team-opal-rents

rents_team-opal.csv · report.html · FEATURES.md · run_everything.py (one command reproduces everything)

Data: Microburbs hackathon files and API (used on the day, not republished) · NSW Spatial Services LiDAR, cadastre, boundaries (CC BY) · OpenStreetMap (ODbL) · NSW DPHI heat and canopy (CC BY 4.0) · ABS Census 2021 (CC BY 4.0) · NSW Fair Trading rental bonds (baseline only)